

Muhsina Tarannum Munfa

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Sylhet Sadar, Sylhet - 3100, Bangladesh

OBJECTIVE

Motivated Computer Science graduate seeking a PhD position in machine learning and natural language processing. Research interests span low-resource language NLP, ML in human computer interaction, and AI in healthcare. Experienced in building NLP pipelines, creating research datasets, and applying transformer-based models to real-world problems.

RESEARCH INTEREST

- Low-Resource and Multilingual NLP – transformer models and cross-lingual transfer for Bengali and underrepresented languages
- AI in Healthcare and HCI – clinical NLP, mental health AI, and explainable human-centered systems

PUBLICATIONS

- Md Sultanul Islam Ovi, **Muhsina Tarannum Munfa**, G.M.M Miftahul Alam Adib, Syed Sabbir Hasan (2026). **A Dual Pipeline Machine Learning Framework for Automated Multi Class Sleep Disorder Screening Using Hybrid Resampling and Ensemble Learning**
[ArXiv:2601.05814](#) | Under review: Social Network Analysis and Mining

SUBMITTED PAPERS

- Author 1, Your Name, Author 3, et al. (Year). **Explainable AI CKD**. *Journal Name*, Vol. XX, Issue X, pp. XXX-XXX. DOI: XX.XXXX/XXXXX.XXXX.XXXXXXX
- Author 1, Your Name, Author 3, et al. (Year). **Suicidal Ideation Dataset**. *Journal Name*, Vol. XX, Issue X, pp. XXX-XXX. DOI: XX.XXXX/XXXXX.XXXX.XXXXXXX

DATASETS

- Author1, Author2 et al (2026). [Dataset Paper Dataset Name]. Associated with manuscript in preparation.
- Bangladesh Climate and Agriculture Dataset** - A curated dataset integrating climate and agricultural variables across Bangladesh districts. 📄
- AQI Dataset For Sylhet, Bangladesh** - Derived from a larger environmental dataset; includes cleaned records, calculated AQI values, and labeled air quality categories for Sylhet region. 📄

EDUCATION

- Metropolitan University** 2018 - 2022
BSc in Computer Science & Engineering City, Country
 - GPA: 3.56/4.00
- GRE** Month Year
Pre-University Education City, Country
 - Grade: XX.X%
- IELTS** Month Year
Secondary Education City, Country
 - GPA: X.X/10

RESEARCH EXPERIENCE

- ResearchDen** 2025 - Present
Research Collaborator Remote
 - Collaborating on healthcare AI and machine learning research projects.
 - Contributing to dataset creation, preprocessing, and experimental design for predictive modeling studies.
 - Assisting in the development and evaluation of explainable AI (XAI) methods for healthcare applications.
 - Participating in literature review, manuscript preparation, and research discussions.
- Prediction of Customers' Purchase Intention in E-commerce Stores** Jan 2022 – Aug 2022
Supervisor: Akhlak Uz Zaman Ashik, Assistant Professor Sylhet, Bangladesh
Undergraduate Thesis Researcher | Department of Computer Science & Engineering, Metropolitan University
 - Investigated customer purchase intention prediction using machine learning techniques on an e-commerce dataset containing 12,330 user sessions.
 - Applied and compared Logistic Regression, K-Nearest Neighbors, Support Vector Machines, Naive Bayes, Decision Tree, Random Forest, and Artificial Neural Networks.

- Performed data preprocessing, including categorical encoding, train-test splitting, and feature scaling.
- Evaluated models using accuracy, precision, recall, F1-score, and confusion matrix analysis.
- Identified Random Forest and Artificial Neural Networks as the highest-performing models for customer purchase prediction.

WORK EXPERIENCE

- OpenGenus IQ** 2022 - 2023
Research Scientist, Intern Remote
 - Contributed technical articles on CS and ML topics to OpenGenus IQ, an open-source knowledge platform used by students and developers globally.
 - Developed ability to explain complex algorithms clearly in written form — directly applicable to technical documentation and knowledge base creation.
- YES Associate** 2023 - 2024
Admission Support Officer Sylhet, Bangladesh
 - Worked collaboratively in a team to support university applicants through each step of the admissions process.
 - Developed strong interpersonal and organizational skills by handling multiple client cases and meeting tight deadlines.

SKILLS

- **Programming Languages:** Python (primary), JavaScript (familiar), ReactJS (developing)
- **Machine Learning:** Scikit-learn, XGBoost, Extra Trees, KNN, Ensemble Methods, SMOTE, LDA, Autoencoders, PyTorch (developing)
- **NLP & Conversational AI:** Intent classification, RAG pipelines, LangChain, ChromaDB, FAISS, STT (BanglaSpeech2Text), Groq API (LLaMA 3), prompt engineering
- **Transformers :** HuggingFace Transformers, DistilBERT, RoBERTa, SentenceTransformers (all-mpnet-base-v2, all-MiniLM-L6-v2), TextBlob, VADER
- **Backend & Data Engineering:** FastAPI, PostgreSQL, MongoDB, REST APIs, Pandas, NumPy, Matplotlib, Seaborn, SQL
- **Deployment & MLOps:** Streamlit, Gradio, Hugging Face Spaces, Render
- **Tools & Platforms:** Git, GitHub, Google Colab, VS Code, Looker Studio, LaTeX, Typst
- **Networking:** Cisco Networking Essentials (CCNA track), Network fundamentals
- **Human Languages:** Bengali (native), English (professional)
- **Research Skills:** Exploratory data analysis, Feature engineering, Academic Writing, Literature Review

PROJECTS

- Bangla Voice Assistant for Low-Literacy Users:**
[A voice assistant breaking smartphone usability barriers among digitally low-literate users] Ongoing Project
Tools: [BanglaSpeech2Text STT model |]
 - Selected BanglaSpeech2Text for server-side speech recognition
 - Lightweight classifier trained on the 1,228-sample dataset
 - Intent-to-document mapping rather than LLM generation
- Air Quality & Health Risk Assessment System:**
[An end-to-end system that monitors AQI, and predicts associated health risk levels] | [Live Demo](#) February - 2026
Tools: FastAPI | PostgreSQL | XGBoost | Streamlit | Looker Studio
 - Built an end-to-end ML system: user inputs a city name → system resolves coordinates from local reference data → calls external weather/pollutant APIs → trained ML model (XGBoost) predicts AQI → result stored in PostgreSQL → dashboard visualizes trends over time
 - Configured PostgreSQL on Neon with schema for time-series AQI storage and managed deployment on Render
 - Built two visualization layers: interactive Streamlit dashboard and Looker Studio dashboard showing AQI trends, health risk distributions, and city-level comparisons
- ChefBot:**
[A RAG-Powered Cooking Assistant] | [Live Demo](#) October - 2025
Tools: [LangChain | ChromaDB | Sentence Transformers | Groq API (LLaMA 3) | Gradio]
 - Built a full RAG pipeline that ingests multiple PDF cookbooks, creates vector embeddings using all-mpnet-base-v2 , and generates responses via LLaMA 3 through Groq API
 - Addressed LLM hallucination by engineering system prompts to strictly constrain responses to retrieved cookbook context only
 - Implemented streaming Gradio interface for responsive UX
- Multi-Model News Sentiment Analysis:**
[Analyzes news sentiment using four different NLP models] | June - 2025
Tools: [HuggingFace Transformers | TextBlob | VADER | DistilBERT | RoBERTa | NewsAPI]

- Designed a comparative NLP pipeline applying four sentiment models (TextBlob, VADER, DistilBERT SST-2, RoBERTa emotion classifier)
- Fetched live news via NewsAPI, storing title, description, and publication date for each article
- Generated two visualization types — label distribution bar charts per model and weekly sentiment score trend lines (showing how sentiment signal fluctuates over time and whether models agree on trend direction)

CERTIFICATIONS

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| • Research Project Lifecycle Workshop
Organized by AI/ML Professional Community Bangladesh | <i>October - 2025</i> |
| • Research Scientist, Intern at OpenGenus | <i>April - 2023</i> |

REFERENCES
